MILITARY STANDARD

MIL-STD-1325-59
(NAVY)

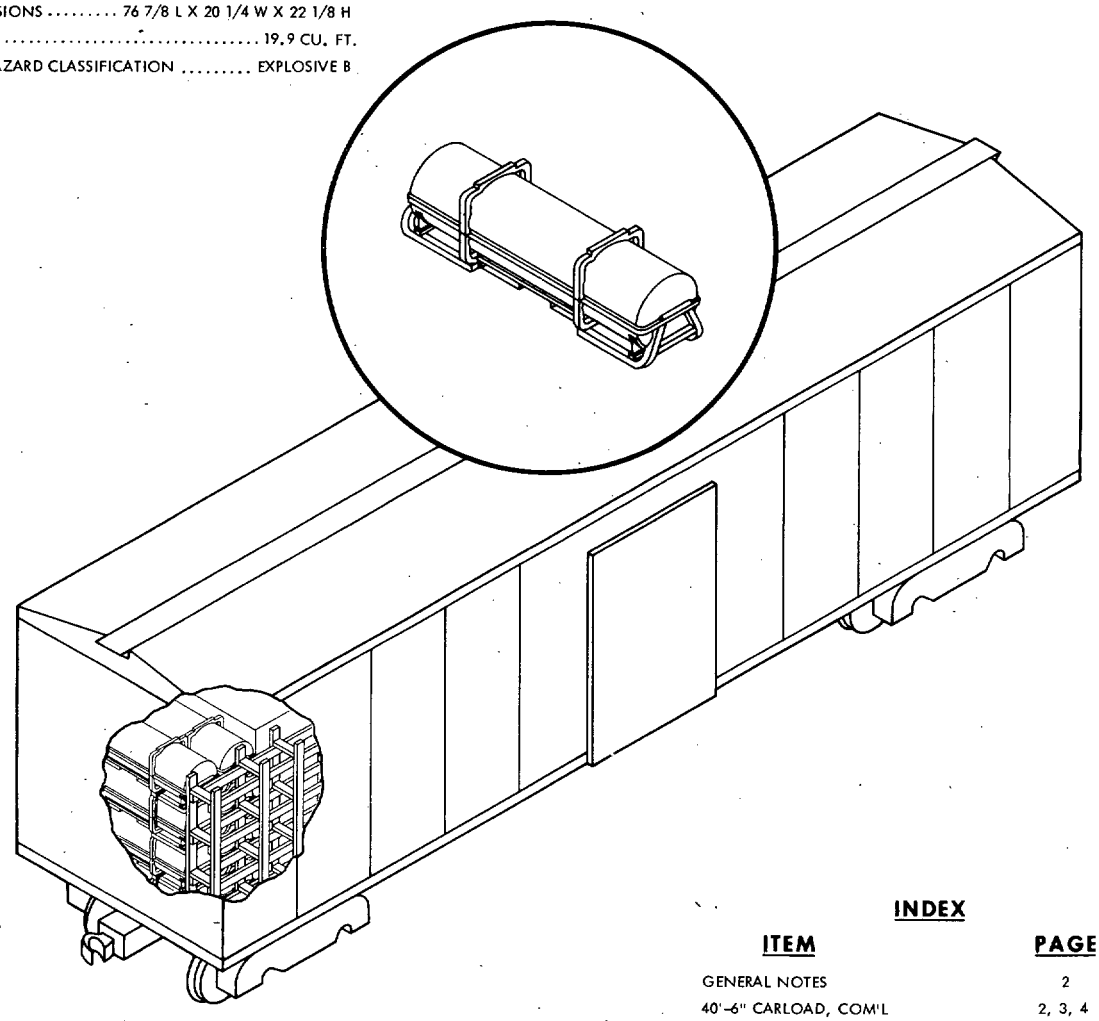
3 AUGUST 1976

SUPERSEDING
WR-52/59 A
10 MAY 1968

RAILCAR LOADING OF HAZARDOUS MATERIALS
ROCKET MOTOR MK 1 MOD 0 OR MK 37 MOD 0 (ASROC) IN CONTAINER MK 178 MOD 0

CONTAINER DATA

DRAWING NO. ..... LD 497698
WEIGHT ..... 607 LBS
DIMENSIONS ..... 76 7/8 L X 20 1/4 W X 22 1/8 H
CUBE ..... 19.9 CU. FT.
DOT HAZARD CLASSIFICATION ..... EXPLOSIVE B



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GENERAL NOTES	2
40'-6" CARLOAD, COM'L	2, 3, 4
50'-6" CARLOAD, COM'L	6,7,8 & 9

NOTE:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HKBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

W.S. Gray NWHC 7/30/76
TECHNICAL DIRECTION AGENT (TDA) DATE

J S Lee 8/4/76
SEASYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck 7/13/76
SIGNATURE SUPERVISOR MILITARY & INTERMODAL SERVICE DATE

ORIGINATOR W. F. Schaul 7/28/76
DATE

NAVAL WEAPONS HANDLING CENTER
WINSTA, EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. DODX UTILITY LOADER CARS SHALL NOT BE USED FOR ITEMS WITH EXPLOSIVES A OR B HAZARD CLASSIFICATION.
3. LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF ASROC ROCKET MOTORS MK 1 MOD 0 AND MK 37 MOD 0 IN MK 178 MOD 0 CONTAINERS. IN ORDER TO EXPEDITE THE HANDLING AND LOADING OPERATIONS, FOUR CONTAINERS MUST BE STACKED AND STRAPPED TOGETHER AS PER THE PROCEDURES SHOWN ON PAGE 9. THE STRAPS MUST BE REMOVED AFTER THE RAILCAR IS UNLOADED.
7. THE CONTAINERS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED SPEC MM-L-751

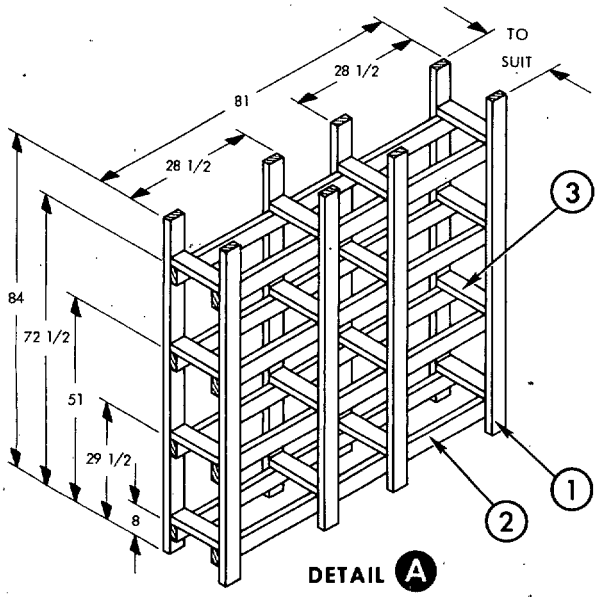
    NAILS - FED SPEC FF-N-105

    STRAPPING - FED SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

    SEALS - FED SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED.

40 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 88 CONTAINERS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325(NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 7 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS LARGER BUT NOT LESS THAN 7 FT WIDE. BOXCARS WITH DOORWAY OPENINGS LESS THAN 7 FT WIDE SHOULD BE REJECTED.

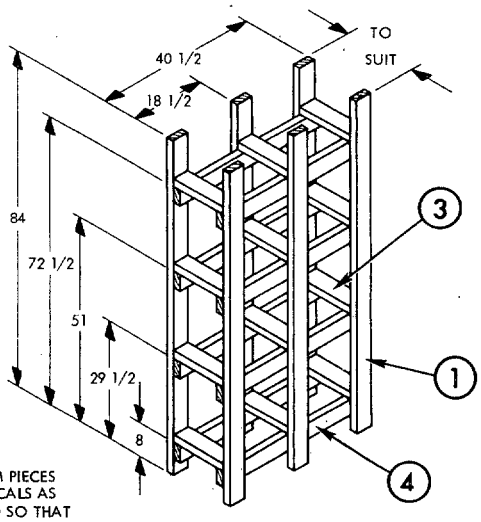


DETAIL A

SWAY BRACING

8 REQUIRED

NOTE:
ADD SHIM PIECES TO VERTICALS AS REQUIRED SO THAT SLACK ACROSS CAR DOES NOT EXCEED ONE INCH.



DETAIL B

SWAY BRACING

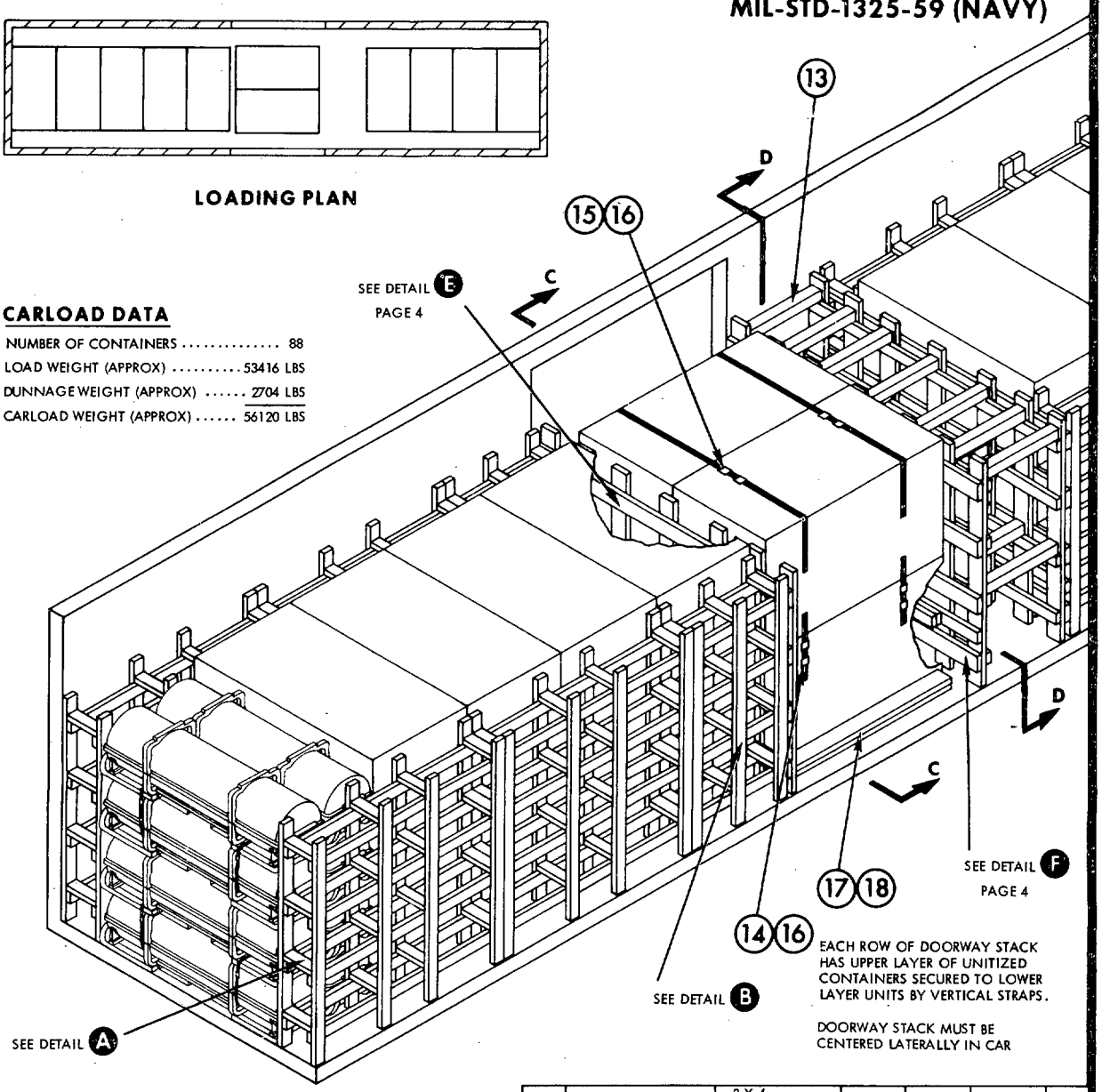
2 REQUIRED

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LOADING PLAN

CARLOAD DATA

NUMBER OF CONTAINERS 88
LOAD WEIGHT (APPROX) ..... 53416 LBS
DUNNAGE WEIGHT (APPROX) ..... 2704 LBS
CARLOAD WEIGHT (APPROX) ..... 56120 LBS



** WHEN BOXCAR HAS NON-NAILABLE FLOOR DOORWAY SWAY BRACE (SEE DETAIL F, PAGE 9) IS REQUIRED IN ADDITION TO DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY).

* 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's

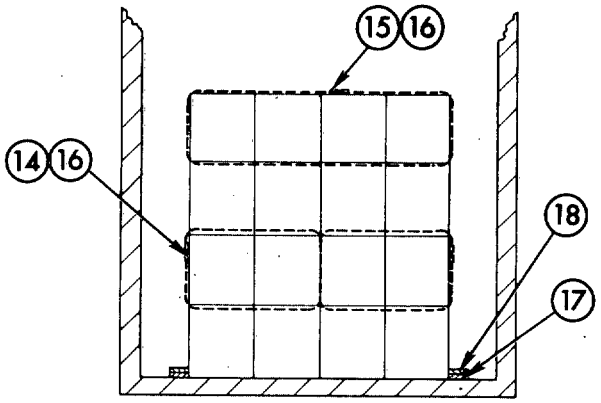
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
18	SLEEPER	2 X 6 X 72	2	17	2 PER FOOT	16d
17	SLEEPER	2 X 6 X 72	2	CAR** FLOOR	2 PER FOOT	16d
16	SEAL	1 1/4	12	-	-	-
15	CROSS STRAP	1 1/4 X .035 X 20 FT 6	2	-	-	-
14	VERTICAL STRAP	1 1/4 X .035 X 12 FT 6	4	-	-	-
13	STRUT	4 X 4 * WEDGE FIT	18	8	2 PER JOINT	16d
12	HOLD DOWN	2 X 4 X TO SUIT	4	11	4	10d

LIST OF MATERIALS & NAILING DATA

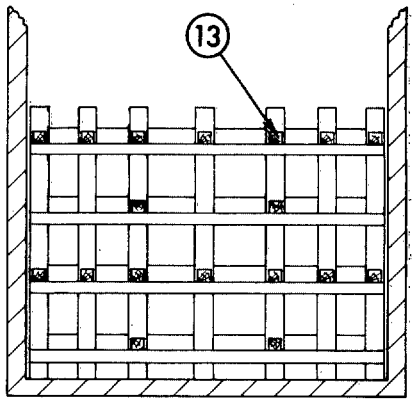
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
11	HOLD DOWN	2 X 4 TO SUIT	4	10	4	10d
10	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-1	2	8	3 PER JOINT	10d
9	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	8	8	3 PER JOINT	10d
8	CENTER GATE VERTICAL	2 X 6 X 84	14	SEE 7, 9	-	-
7	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	8	8	3 PER JOINT	10d
6	SEPARATOR GATE VERTICAL	2 X 6 X 84	7	SEE 5	-	-
5	SEPARATOR GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	4	6	3 PER JOINT	10d
4	SWAY BRACE HORIZONTAL	2 X 4 X 40 1/2	16	1	3 PER JOINT	10d
3	SWAY BRACE STRUT	2 X 4 X TO SUIT	152	1	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 81	64	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 84	76	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA

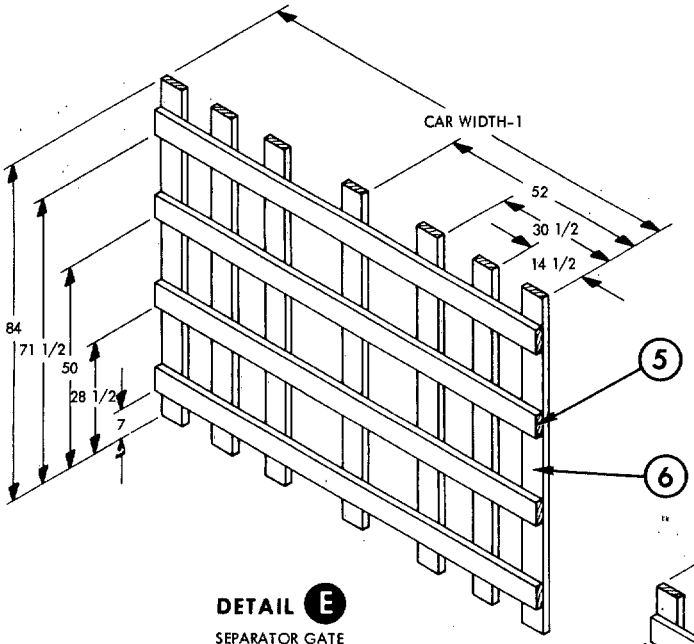
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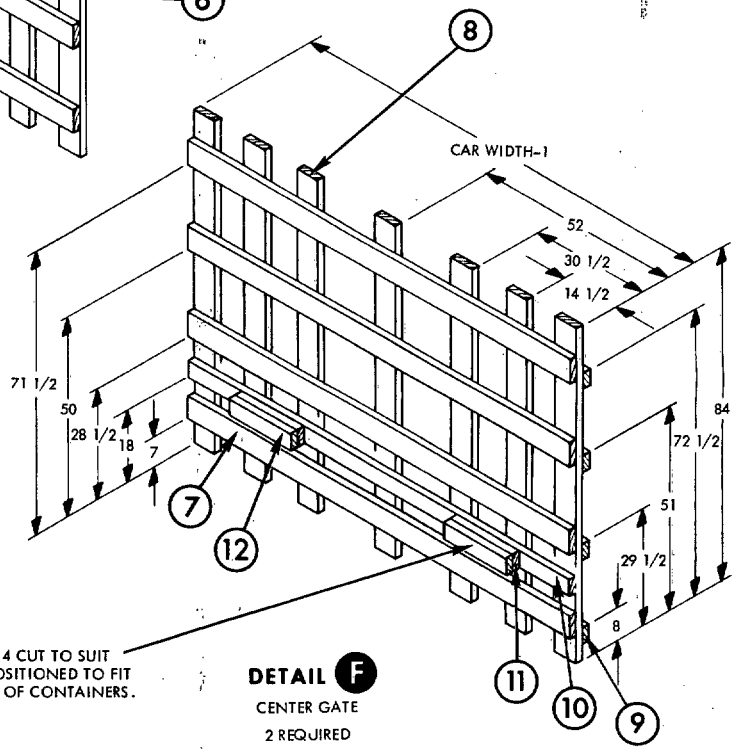
SECTION C - C
SHOWING LOCATION OF STRAPS



SECTION D - D
SHOWING LOCATION OF STRUTS



DETAIL E
SEPARATOR GATE
1 REQUIRED



DETAIL F
CENTER GATE
2 REQUIRED

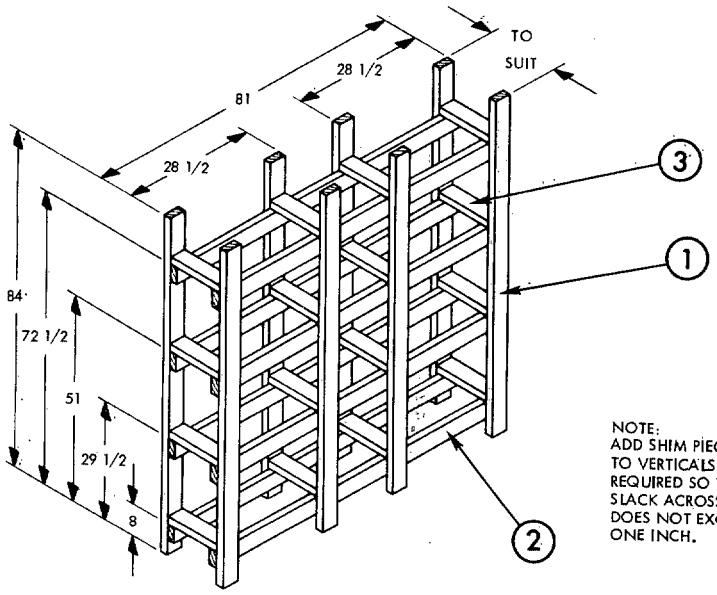
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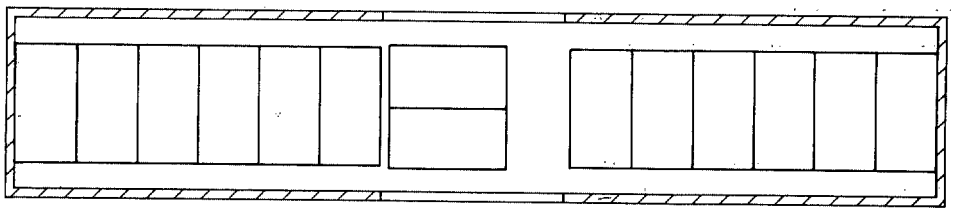
50 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 112 CONTAINERS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325(NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE BUT NOT LESS THAN 7 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS. PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



NOTE:
ADD SHIM PIECES TO VERTICALS AS REQUIRED SO THAT SLACK ACROSS CAR DOES NOT EXCEED ONE INCH.

DETAIL A
SWAY BRACING
12 REQUIRED

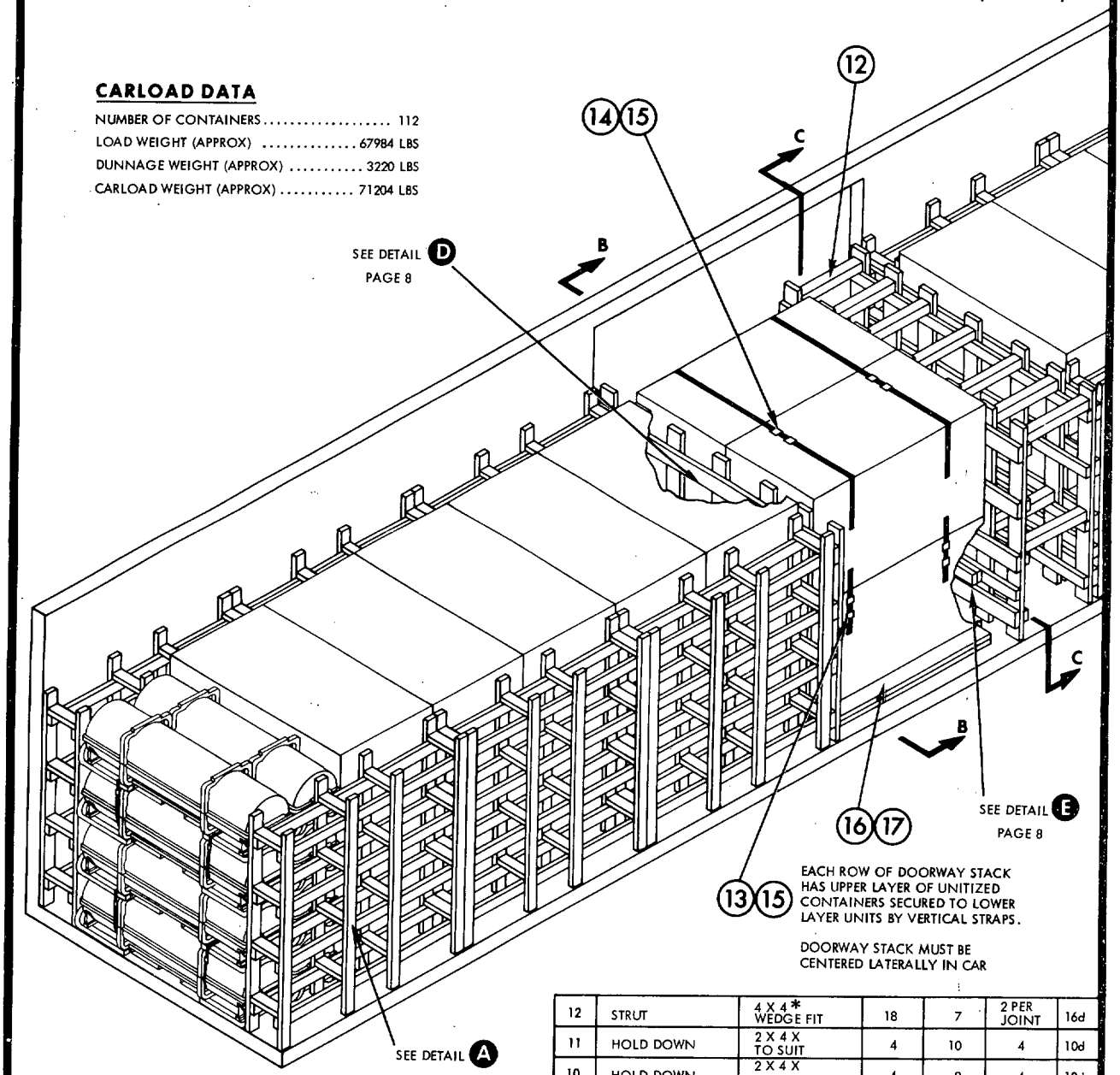


LOADING PLAN

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CARLOAD DATA

NUMBER OF CONTAINERS 112
LOAD WEIGHT (APPROX) ..... 67984 LBS
DUNNAGE WEIGHT (APPROX) ..... 3220 LBS
CARLOAD WEIGHT (APPROX) ..... 71204 LBS



** WHEN BOX CAR HAS NON-NAILABLE FLOOR DOORWAY SWAY BRACE (SEE DETAIL F, PAGE 9) IS REQUIRED IN ADDITION TO DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY).

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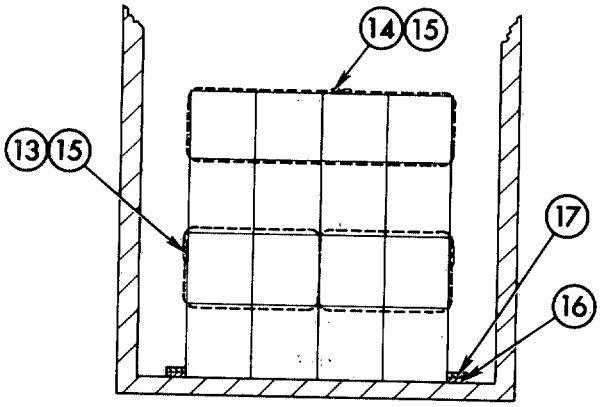
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
17	SLEEPER	2 X 6 X 72	2	16	2 PER FOOT	16d
16	SLEEPER	2 X 6 X 72	2	CAR** FLOOR	2 PER FOOT	16d
15	SEAL	1 1/4	12	-	-	-
14	CROSS STRAP	1 1/4 X .035 X 20 FT 6	2	-	-	-
13	VERTICAL STRAP	1 1/4 X .035 X 12 FT 6	4	-	-	-

LIST OF MATERIALS & NAILING DATA

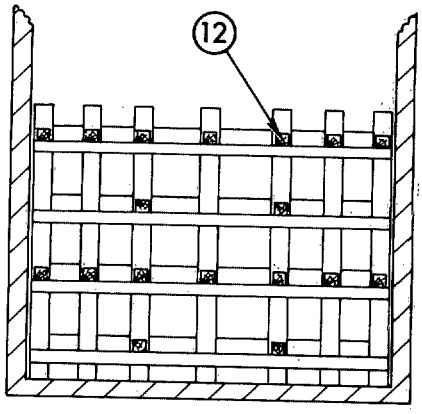
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
12	STRUT	4 X 4* WEDGE FIT	18	7	2 PER JOINT	16d
11	HOLD DOWN	2 X 4 X TO SUIT	4	10	4	10d
10	HOLD DOWN	2 X 4 X TO SUIT	4	9	4	10d
9	HOLD DOWN SPACER	2 X 4 X CAR WIDTH-1	2	7	3 PER JOINT	10d
8	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	8	7	3 PER JOINT	10d
7	CENTER GATE VERTICAL	2 X 6 X 84	14	SEE 6, 8	-	-
6	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	8	7	3 PER JOINT	10d
5	SEPARATOR GATE VERTICAL	2 X 6 X 84	7	SEE 4	-	-
4	SEPARATOR GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	4	5	3 PER JOINT	10d
3	SWAY BRACE STRUT	2 X 4 X TO SUIT	192	1	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 81	96	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 84	96	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA

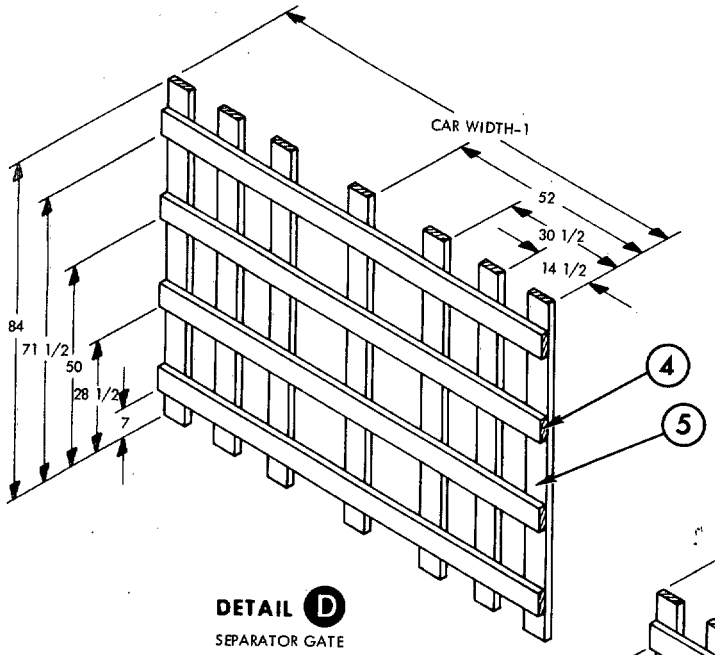
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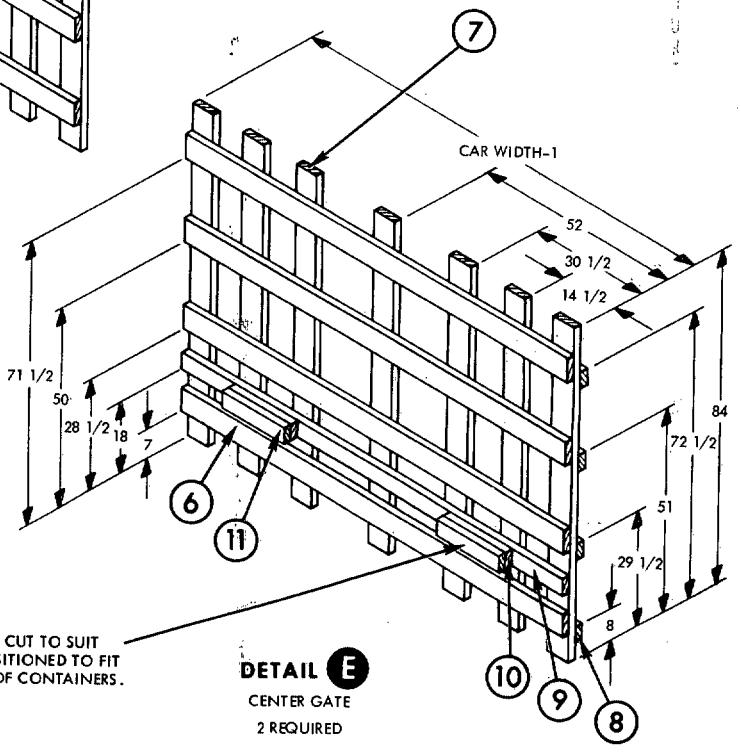
SECTION B - B
SHOWING LOCATION OF STRAPPING



SECTION C - C
SHOWING LOCATION OF STRUTS



DETAIL D
SEPARATOR GATE
1 REQUIRED

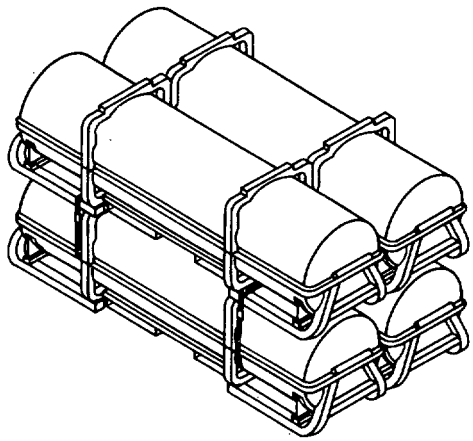


DOUBLED 2 X 4 CUT TO SUIT AND FIELD POSITIONED TO FIT UNDER SKIDS OF CONTAINERS.

DETAIL E
CENTER GATE
2 REQUIRED

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STACKING AND STRAPPING FOR CARLOADING ONLY



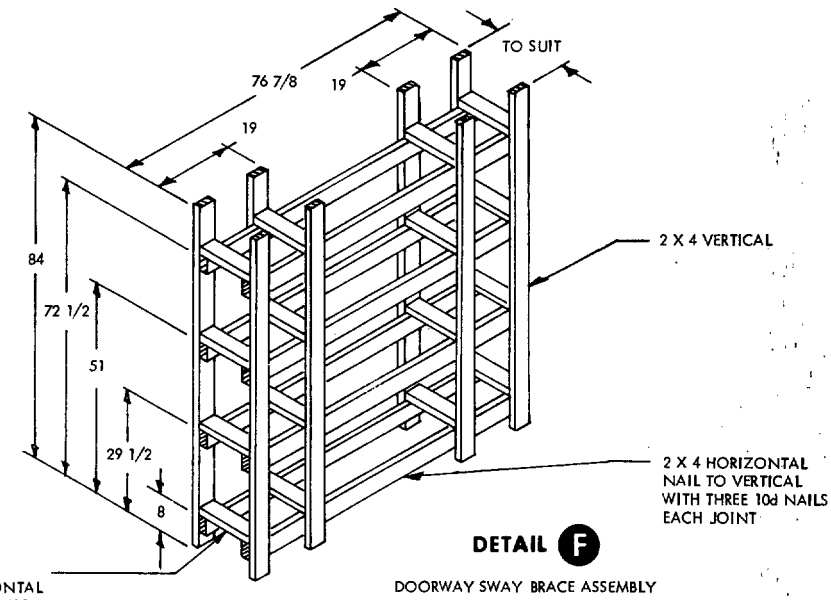
TO EXPEDITE CARLOADING HANDLING AND LOADING OPERATIONS, FOUR CONTAINERS MUST BE STACKED AND STRAPPED TOGETHER IN ACCORDANCE WITH THE FOLLOWING PROCEDURES:

1. POSITION BOTTOM CONTAINERS SIDE BY SIDE WITH FORE AND AFT ENDS IN THE SAME DIRECTION.
2. STACK A TOP CONTAINER ON EACH BOTTOM ONE WITH FORE AND AFT END IN THE SAME DIRECTION AS THE BOTTOM CONTAINER.

NOTE

ALL STOCKING FEATURES MUST BE IN ALIGNMENT

3. POSITION TWO 3/4" X .035 X 13 FT STEEL STRAPS UNDER BODY OF BOTTOM TWO CONTAINERS AND OVER SKIDS OF TOP TWO CONTAINERS AS SHOWN. TENSION AND SEAL WITH TWO DOUBLE CRIMPED 3/4" SEALS.
4. REMOVE STRAPPING FROM CONTAINERS AFTER UNLOADING THE BOXCAR.



DETAIL F

DOORWAY SWAY BRACE ASSEMBLY FOR NON-NAILABLE FLOORS

REVIEW ACTIVITY
NAVY OS

PREPARING ACTIVITY
NAVY OS
(PROJECT NO. 8140-N331)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

Mil-Std-1325-59 (NAVY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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